

UNIT 9

THE SOLAR SYSTEM

Learning objectives:

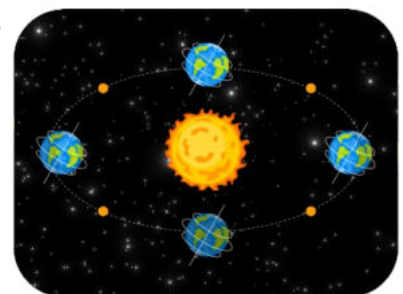
- Learn about the Sun, planets, moons, asteroids, and comets in the Solar System.
- Identify and describe the eight planets and their unique features.
- Understand the role of the Sun in providing heat and light for life on Earth.
- Explore other objects in the Solar System, like moons and asteroids.
- Apply knowledge by creating a Solar System model.
- Reflect on how the Solar System impacts life on Earth.

The Solar System is our cosmic neighbourhood! It includes the **Sun**, the **planets**, and other objects like **moons, asteroids, and comets**. Our planet, Earth, is one of eight planets that orbit, or travel around, the Sun. Each part of the Solar System plays a unique role, and it's fascinating to learn how they all move and interact in space.

THE SUN

The **Sun** is the centre of the Solar System and the biggest object in it! It's a giant ball of hot, glowing gases that gives off **light and heat**. This light and heat are essential for life on Earth,

providing energy for plants, warmth for all living things, and light so we can see. The Sun's gravity is what keeps all the planets in the Solar System in their orbits, moving around it.



THE PLANETS

There are **eight planets** in our Solar System, each unique and amazing in its own way. Let's take a quick tour of each planet, starting from the one closest to the Sun.

1. Mercury

- Mercury is the closest planet to the Sun and is very hot during the day but extremely cold at night.
- It's also the smallest planet in the Solar System.

2. Venus

- Venus is the brightest planet and can often be seen in the morning or evening sky. It's also very hot, even hotter than Mercury!
- Venus has thick clouds that trap heat, making it the hottest planet.

3. Earth

- Earth is our home and the only planet we know of that has life. It has water, air, and perfect temperatures for living things.
- Earth has one moon, which we see shining in the night sky.

4. Mars

- Mars is called the "Red Planet" because of its reddish colour, which comes from iron-rich soil.
- Scientists are studying Mars closely to see if life ever existed there in the past.



5. Jupiter

- Jupiter is the largest planet in the Solar System. It's so big that over 1,300 Earths could fit inside it!
- Jupiter has a huge storm called the Great Red Spot, which is larger than Earth and has been going on for hundreds of years.

6. Saturn

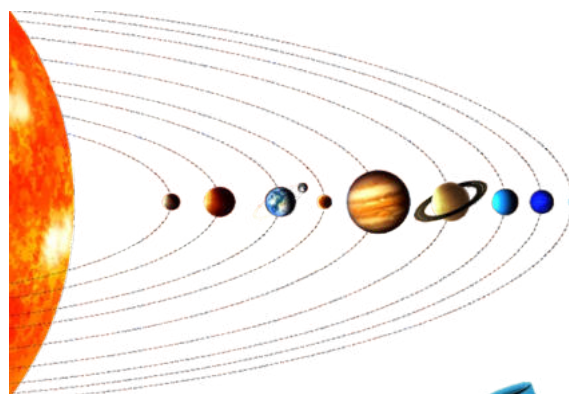
- Saturn is known for its stunning rings, which are made up of ice and rock particles.
- It's the second-largest planet and has many moons orbiting it.

7. Uranus

- Uranus is unique because it spins on its side, unlike any other planet in the Solar System.
- It has a pale blue colour because of the gases in its atmosphere.

8. Neptune

- Neptune is the farthest planet from the Sun and is very cold and windy.
- It has a deep blue colour and also has powerful storms in its atmosphere.



MOONS, ASTEROIDS, AND COMETS

Besides planets, the Solar System has other interesting objects that orbit the Sun.

Moons

Most planets have moons that orbit around them. Moons are like mini-worlds that travel with their planets. Earth has one moon, but other planets like Jupiter and Saturn have many moons—Jupiter has over 75!



Asteroids

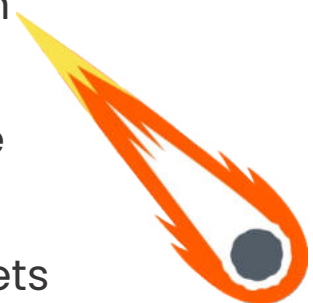
Asteroids are small, rocky objects that orbit the Sun, mostly found in the **Asteroid Belt** between Mars and Jupiter.



Some asteroids are as small as pebbles, while others are hundreds of kilometers wide.

Comets

Comets are icy objects that orbit the Sun in long, stretched-out paths. When they get close to the Sun, they heat up, and their ice turns into gas, forming a beautiful tail that can stretch for millions of kilometers. Comets are often called “space snowballs” because they are made of ice and dust.



Why is the Solar System Important?

Learning about the Solar System helps us understand our place in space. The Sun gives us light and warmth, the planets show us how diverse worlds can be, and moons, asteroids, and comets add to the wonder of our universe. Scientists study the Solar System to learn about Earth's history, the possibility of life on other planets, and how to protect our planet from things like asteroid impacts.

Fun Facts to Remember

Fact 1

The planets closest to the Sun (Mercury, Venus, Earth, and Mars) are called the **Inner Planets** and are rocky. The ones farther away (Jupiter, Saturn, Uranus, and Neptune) are called the **Outer Planets** and are mostly made of gas.

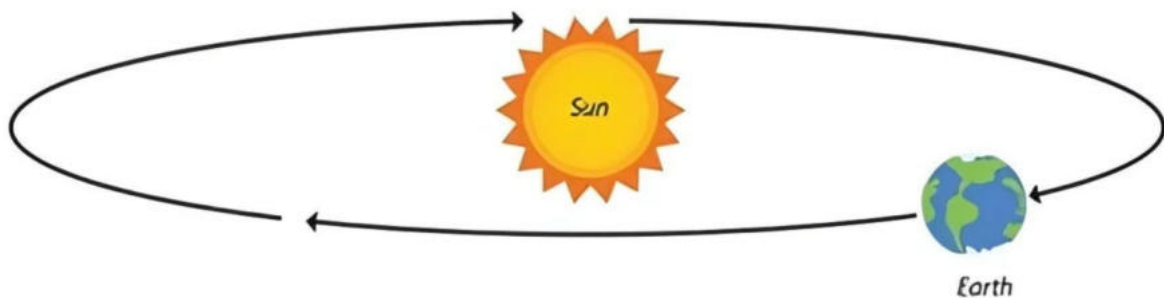
Fact 2

A day on Jupiter is just about 10 hours long, while a day on Venus is longer than a year on Earth!



Fill in the blanks

1. The Solar System is made up of the Sun, the planets, and other objects like _____, _____ and _____
2. The Sun provides us with _____ and _____ which are essential for life on Earth.
3. The planet closest to the Sun is _____ .
4. _____ is known as the "Red Planet."
5. The largest planet in the Solar System is _____.
6. _____ has beautiful rings made of ice and rock particles.



Write "T" for True or "F" for False for each statement.

- The Sun is the centre of the Solar System.
- Venus is the only planet in the Solar System with life.
- Jupiter is smaller than Earth.
- Earth has two moons.
- Comets are made of ice and dust and form a tail when they get close to the Sun.
- Neptune is the farthest planet from the Sun.



Multiple Choice Questions .

1. What is the main source of heat and light for our Solar System?

- Jupiter
- The Sun
- Mars
- The Moon

2. Which planet is known for its rings?

- Mars
- Earth
- Saturn
- Venus

3. Which planet is called the “Red Planet”?

- Mercury
- Venus
- Mars
- Neptune

4. Which of these is the largest planet in the Solar System?

- Earth
- Mars
- Jupiter
- Saturn



5. Which planet is closest to the Sun?

- Earth
- Neptune
- Mercury
- Venus

6. What are asteroids mostly made of?

- Ice and dust
- Gas
- Rocks
- Metal

Fun Fact

The Sun is so big that you could fit over 1 million Earths inside it!



Answer the following questions.

Q1: Name the eight planets in the Solar System in order from the Sun?

Q2: What makes Earth different from other planets in the Solar System?

Q3: Describe what a comet is and what happens when it comes close to the Sun?

Q4: Why is the Sun important for life on Earth?

Q5: Which planet has a storm called the Great Red Spot?

Q6: What is an asteroid, and where are most asteroids found in the Solar System?

Fun Fact

The Solar System is part of a galaxy called the **Milky Way**, which contains billions of stars!





Make Your Own Solar System Model



ACTIVITY

Make your own solar system model to learn about the planets in the solar system and their order from the sun.

Materials Needed

Coloured paper or play dough

Markers or crayons

String

Glue or tape



Instructions:

1. Use coloured paper or play dough to create small models of each planet in the Solar System, making sure they differ in size (e.g., Jupiter should be bigger than Earth).
2. Label each planet with its name.
3. Arrange the planets in order from the Sun, attaching them to a piece of string or on a poster board.
4. Present your model to your classmates, explaining one interesting fact about each planet.

